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# Inference in belief networks

## Chapter 15.3–4 + new

# Outline

◇ Exact inference by enumeration

◇ Exact inference by variable elimination

◇ Approximate inference by stochastic simulation

◇ Approximate inference by Markov chain Monte Carlo

# Inference tasks

Šimple queries: compute posterior marginal $\mathbf{P}(X_i|\mathbf{E}=\mathbf{e})$
e.g., $P(NoGas|Gauge=empty, Lights=on, Starts=fa$

Čonjunctive queries: $\mathbf{P}(X_i, X_j|\mathbf{E}=\mathbf{e}) = \mathbf{P}(X_i|\mathbf{E}=\mathbf{e})\mathbf{P}(X_j|X_i$

Ŏptimal decisions: decision networks include utility informatio
probabilistic inference required for $P(outcome|action, e$

V̌alue of information: which evidence to seek next?

Šensitivity analysis: which probability values are most critical?

Ĕxplanation: why do I need a new starter motor?

# Inference by enumeration

Slightly intelligent way to sum out variables from the joint witho
constructing its explicit representation

Simple query on the burglary network:
$\mathbf{P}(B|J = true, M = true)$
$= \mathbf{P}(B, J = true, M = true)/P(J = true, M = true)$
$= \alpha\mathbf{P}(B, J = true, M = true)$
$= \alpha\Sigma_e\Sigma_a\mathbf{P}(B, e, a, J = true, M = true)$

Rewrite full joint entries using product of CPT entries:
$P(B = true|J = true, M = true)$
$= \alpha\Sigma_e\Sigma_a P(B = true)P(e)P(a|B = true, e)P(J = true|a)P($
$= \alpha P(B = true)\Sigma_e P(e)\Sigma_a P(a|B = true, e)P(J = true|a)P($

# Enumeration algorithm

Exhaustive depth-first enumeration: $O(n)$ space, $O(d^n)$ time

ENUMERATIONASK(X̌,**e**,b̌n) **returns** a distribution over X̌
**inputs**: X̌, the query variable
**e**, evidence specified as an event
b̌n, a belief network specifying joint distribution $\mathbf{P}(X_1, \ldots, X_n)$

$\mathbf{Q}(X)$ ← a distribution over X̌
**for each** value $x_i$ of X̌ **do**
extend **e** with value $x_i$ for X̌
$\mathbf{Q}(x_i)$ ← ENUMERATEALL(VARS[b̌n],**e**)
**return** NORMALIZE($\mathbf{Q}(X)$)

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ENUMERATEALL(v̌ars,**e**) **returns** a real number
**if** EMPTY?(v̌ars) **then return** 1.0
**else do**
Y̌ ← FIRST(v̌ars)
**if** Y̌ has value y̌ in **e**
**then return** $P(y|Pa(Y)) \times$ ENUMERATEALL(REST(v̌ars),**e**)
**else return** $\sum_y P(y|Pa(Y)) \times$ ENUMERATEALL(REST(v̌ars),$\mathbf{e}_y$)
where $\mathbf{e}_y$ is **e** extended with $Y = y$

# Inference by variable elimination

Enumeration is inefficient: repeated computation
e.g., computes $P(J = true|a)P(M = true|a)$ for each value

Variable elimination: carry out summations right-to-left,
storing intermediate results (factors) to avoid recomputation

$$
\begin{aligned}
\mathbf{P}(B|J = true, M = true) \\
&= \alpha \underbrace{\mathbf{P}(B)}_{B} \textstyle\sum_e \underbrace{P(e)}_{E} \textstyle\sum_a \underbrace{\mathbf{P}(a|B,e)}_{A} \underbrace{P(J = true|a)}_{J} \underbrace{P(M = t}_{M} \\
&= \alpha \mathbf{P}(B) \textstyle\sum_e P(e) \textstyle\sum_a \mathbf{P}(a|B,e) P(J = true|a) f_M(a) \\
&= \alpha \mathbf{P}(B) \textstyle\sum_e P(e) \textstyle\sum_a \mathbf{P}(a|B,e) f_J(a) f_M(a) \\
&= \alpha \mathbf{P}(B) \textstyle\sum_e P(e) \textstyle\sum_a f_A(a,b,e) f_J(a) f_M(a) \\
&= \alpha \mathbf{P}(B) \textstyle\sum_e P(e) f_{\bar{A}JM}(b,e) \text{ (sum out } A) \\
&= \alpha \mathbf{P}(B) f_{\bar{E}\bar{A}JM}(b) \text{ (sum out } E) \\
&= \alpha f_B(b) \times f_{\bar{E}\bar{A}JM}(b)
\end{aligned}
$$

# Variable elimination: Basic operation

Pointwise product of factors $f_1$ and $f_2$:

$$f_1(x_1, \ldots, x_j, y_1, \ldots, y_k) \times f_2(y_1, \ldots, y_k, z_1, \ldots, z_l)$$
$$= f(x_1, \ldots, x_j, y_1, \ldots, y_k, z_1, \ldots, z_l)$$

E.g., $f_1(a, b) \times f_2(b, c) = f(a, b, c)$

Summing out a variable from a product of factors: move an
factors outside the summation:

$$\Sigma_x f_1 \times \cdots \times f_k = f_1 \times \cdots \times f_i \Sigma_x f_{i+1} \times \cdots \times f_k = f_1 \times \cdots$$

assuming $f_1, \ldots, f_i$ do not depend on $X$

# Variable elimination algorithm

**function** ELIMINATIONASK(X,**e**,bn) **returns** a distribution over X
**inputs**: $X$, the query variable
    **e**, evidence specified as an event
    *bn*, a belief network specifying joint distribution $\mathbf{P}(X_1, \ldots, X_n)$

**if** $X \in \mathbf{e}$ **then return** observed point distribution for X
factors ← [ ]; vars ← REVERSE(VARS[bn])
**for each** var **in** vars **do**
factors ← [MAKEFACTOR($var$, **e**)|$factors$]
**if** var is a hidden variable **then** factors ← SUMOUT(var,factors)
**return** NORMALIZE(POINTWISEPRODUCT(factors))

# Complexity of exact inference

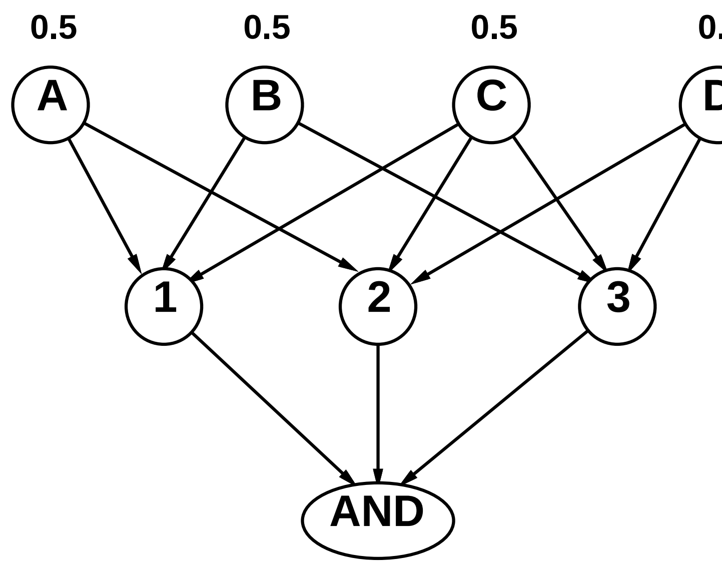
Singly connected networks (or polytrees):
– any two nodes are connected by at most one (undirected)
– time and space cost of variable elimination are $O(d^k n)$

Multiply connected networks:
– can reduce 3SAT to exact inference $\Rightarrow$ NP-hard
– equivalent to *counting* 3SAT models $\Rightarrow$ #P-complete

**1. A v B v C**

**2. C v D v ~A**

**3. B v C v ~D**



# Inference by stochastic simulation

Basic idea:

1) Draw $N$ samples from a sampling distribution $S$
2) Compute an approximate posterior probability $\hat{P}$
3) Show this converges to the true probability $P$

Outline:

– Sampling from an empty network
– Rejection sampling: reject samples disagreeing with evide
– Likelihood weighting: use evidence to weight samples
– MCMC: sample from a stochastic process whose stationar
distribution is the true posterior